

WHEATSTONE BRIDGE (08)**AIM:**

Investigation of the temperature variation of a bulb filament with current and verification of the Stefan's law of radiation.

APPARATUS:

6 V/ 0.4 A bulb with holder, two 5 k Ω resistance boxes, one 500 Ω resistance box, one 5 Ω resistance box, a power supply, a rheostat, a plug key, a tap key, a multi meter, an ammeter, a center-zero galvanometer, and a stand.

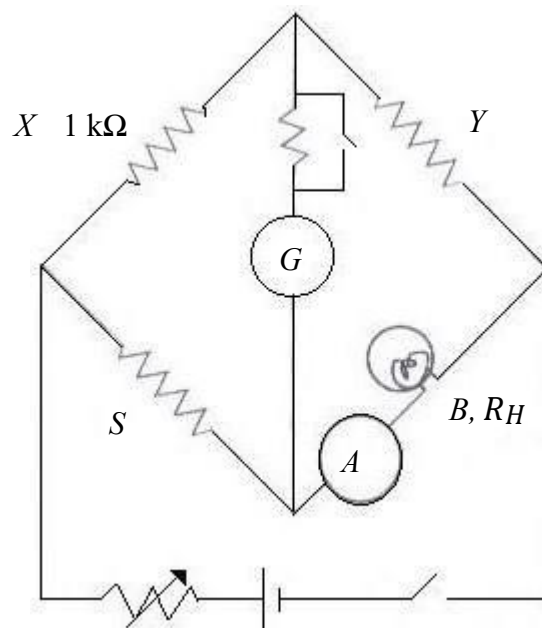
THEORY:

Fig. 1: The figure of the experimental circuit.

Where, X, Y, S : Resistor Boxes, B : Bulb, R_H : Resistance of the bulb A : Ammeter, G : Center zero galvanometer

At the balance of the Fig. 1;

$$\frac{X}{Y} = \frac{S}{R_H} \quad (1)$$

At a given temperature of the filament, resistance can be given by,

$$R(t) = R(t_0)[1 + \alpha(t - t_0)] \quad (2)$$

where,

α : temperature coefficient of the filament material (Tungsten)

$R(t)$: resistance at temperature t

$R(t_0)$: resistance at room temperature

t : room temperature

WHEATSTONE BRIDGE (08)

Power dissipated by the bulb;

$$P = I^2 R(t) \quad (3)$$

By Stefan's law;

$$P = \varepsilon A \sigma T^4 \quad (4)$$

where,

ε : emissivity of the surface of the filament

A : total surface area of the filament

σ : Stefan's constant

T : Temperature in Kelvin

By (3) and (4);

$$I^2 R(t) = \varepsilon A \sigma T^4 \quad (5)$$

By taking natural log on both sides of the equation (5);

$$\ln(I^2 R(t)) = 4 \ln(T) + \ln(\varepsilon A \sigma) \quad (6)$$

WHEATSTONE BRIDGE (08)**PROCEDURE:**

- Measure the resistance of the bulb at room temperature.
- Set up the Wheatstone bridge circuit with the bulb (B) in one arm as shown in the Fig. 1.
- Adjust the rheostat to obtain the desired current I through the bulb.
- Balance the bridge circuit by varying Y and note down the value of the balanced Y for current I .
- Repeat the above procedure for different values of current for a chosen range.
- Calculate the resistance R of the filament for different values of current I using the equation (1).
- Calculate the temperature of the filament for different values of current I using the equation (2).
- Calculate the power dissipated by the bulb at different temperatures using $I^2R(t)$.
- Plot the graph of $\ln(I^2R(t))$ vs. $\ln(T)$ (in Kelvin) and find the gradient.
- Compare the theoretical value of the gradient the experimental value of the gradient to verify the Stefan's law of radiation.